EV BATTERY MANAGEMENT SYSTEMS — TECHNICAL REPORT

State of Charge Estimation for Electric Vehicle BMS

Coulomb Counting vs. Extended Kalman Filter on a 1RC Thévenin-Equivalent Cell Model

AUTHOR	Vidhanshu Tenwalia

CELL UNDER TEST	Panasonic 18650PF • 2.9 Ah nominal
VALIDATION CYCLES	UDDS (urban / gentle) • US06 (aggressive)
TOOLCHAIN	MATLAB / Simulink
PORTFOLIO	github.com/vidhanshu-tenwalia

HEADLINE RESULT

Joint-tuned EKF holds SOC error to 0.93–0.95% across both drive cycles — a 4.1x reduction over Coulomb Counting on UDDS — versus Coulomb Counting's 0.79–3.81% swing.

01 Executive Summary

Accurate State of Charge (SOC) estimation is one of the core functions of an Electric Vehicle (EV) Battery Management System (BMS): it underpins safety interlocks, range prediction, and charge/discharge power limits. This report documents the design, tuning, and validation of an Extended Kalman Filter (EKF) SOC estimator built on a 1RC Thévenin equivalent circuit model, benchmarked against a standard Coulomb Counting estimator under both a gentle urban drive cycle (UDDS) and an aggressive highway/acceleration cycle (US06).

The central finding is that a **jointly-tuned** EKF — tuned to minimize worst-case error across both cycles rather than optimized for a single one — delivers tightly bounded SOC error (0.93–0.95% RMSE) regardless of driving style, while Coulomb Counting's accuracy swings nearly 5x (0.79–3.81% RMSE) depending on the cycle and degrades without bound as current-sensor drift accumulates over time.

4.1x	0.93–0.95%	2
Lower RMSE vs. Coulomb Counting on UDDS	EKF error band across both drive cycles	Drive cycles jointly optimized (not overfit)

02 System Architecture and Workflow

The full pipeline moves from raw cell characterization data through model construction to a head-to-head estimator comparison. Each stage feeds the next; the same battery model and the same degraded current/voltage signals are used for both estimators, so the RMSE comparison in Section 7 isolates the estimation algorithm as the only variable.

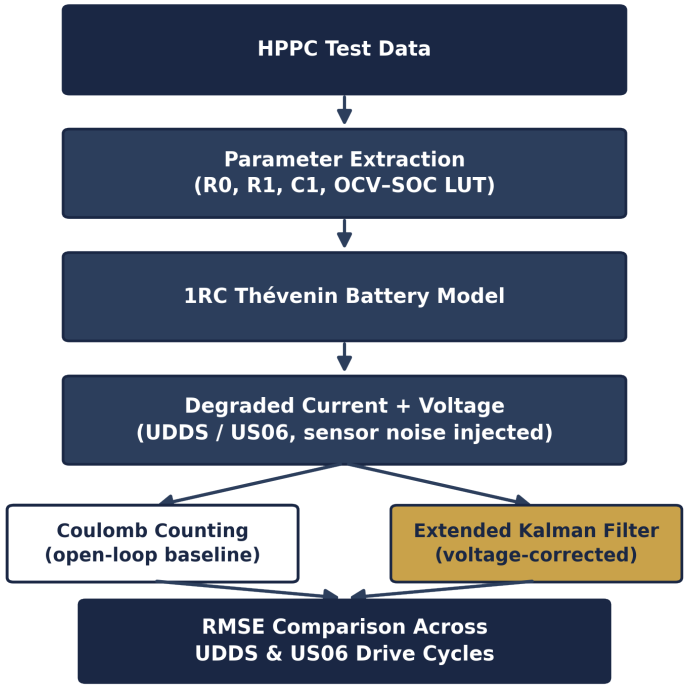

Figure 1. End-to-end workflow: from HPPC characterization to final RMSE comparison.

03 System Modeling and Parameter Extraction

The EKF's accuracy depends entirely on how well its internal model reflects true cell dynamics. A 1RC Thévenin equivalent circuit was constructed for a Panasonic 18650PF cell (nominal capacity Q_{rated} = 2.9 Ah), with all parameters extracted from Hybrid Pulse Power Characterization (HPPC) test data rather than assumed:

- **Ohmic resistance (R0) = 25.5 mΩ** — extracted from the instantaneous voltage step at the onset of each current pulse.
- **Polarization branch (R1 = 4.5 mΩ, C1 = 2795.4 F)** — derived by exponential curve-fitting the post-pulse voltage relaxation.
- **OCV–SOC relationship** — a 15-point lookup table built from fully relaxed open-circuit voltage checkpoints across the full SOC range.

Parameter	Symbol	Value	Extraction Method
Nominal capacity	Q_rated	2.9 Ah	Manufacturer datasheet
Ohmic resistance	R0	25.5 mΩ	HPPC instantaneous step
Polarization resistance	R1	4.5 mΩ	HPPC relaxation curve fit
Polarization capacitance	C1	2795.4 F	HPPC relaxation curve fit
OCV–SOC map	OCV(SOC)	15-point LUT	Relaxed voltage checkpoints

Table 1. 1RC Thévenin model parameters for the Panasonic 18650PF cell.

Equivalent Circuit

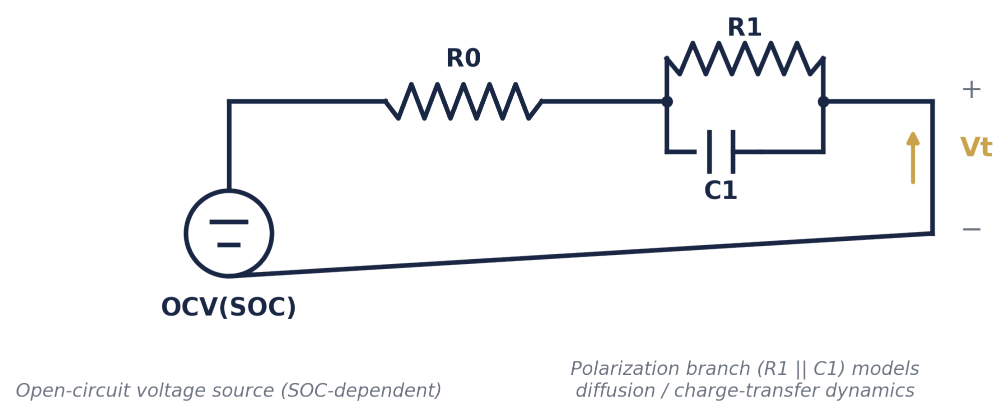

Figure 2. 1RC Thévenin equivalent circuit: an SOC-dependent OCV source, series ohmic resistance R0, and a parallel R1–C1 branch capturing polarization dynamics.

HPPC Parameter Extraction

R0 is read from the instantaneous voltage step at pulse onset/offset (median of 67 pulses); R1 and $\tau = R1 \cdot C1$ are obtained by fitting the exponential relaxation that follows (median of 13 clean relaxation curves). The reconstruction below uses these actual fitted values (R0 = 25.5 mΩ, R1 = 4.5 mΩ, τ = 12.52 s); the current pulse magnitude is representative of the dataset's pulse amplitude, since a single logged pulse trace was not exported separately.

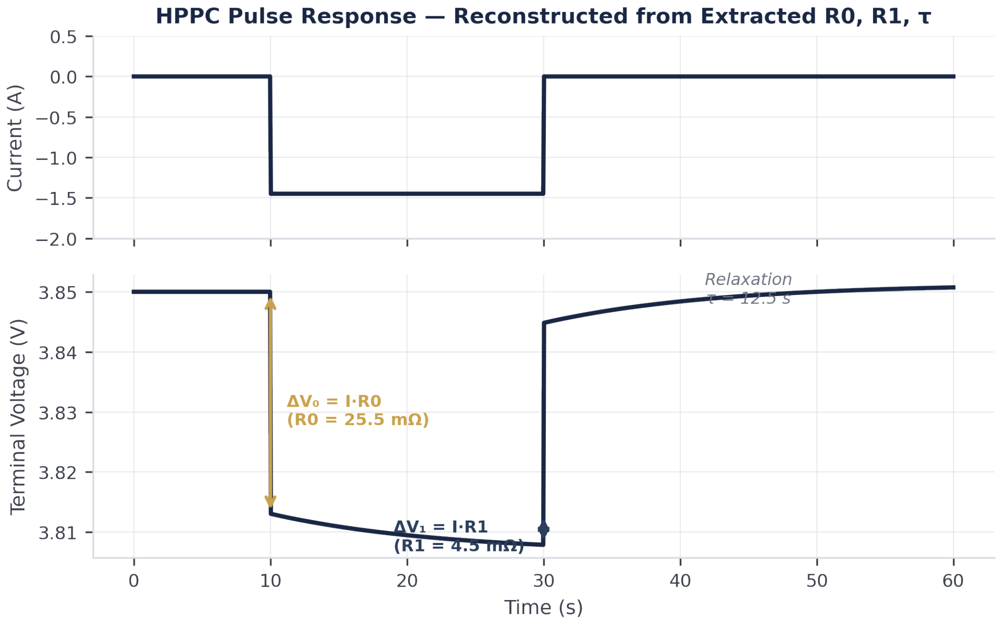

Figure 3. HPPC pulse-response reconstruction using the extracted R0, R1, and τ parameters, showing the instantaneous-drop / relaxation segmentation used for extraction.

OCV–SOC Characteristic

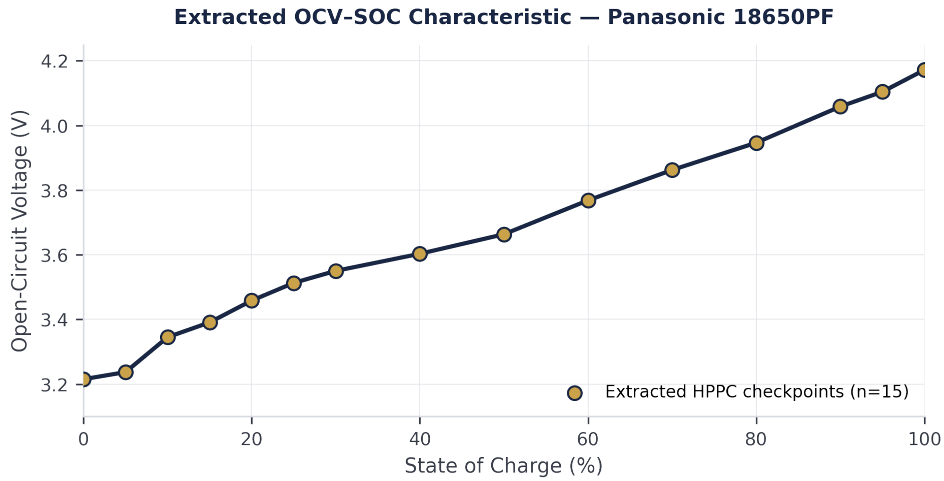

Figure 4. Extracted OCV–SOC curve — 15 checkpoints from 66 fully-relaxed rest periods (>1000 s), snapped to standard 5% SOC targets.

04 Estimation Methodology

4.1 Drive Cycle Preparation

To reproduce realistic sensor imperfection rather than idealized lab conditions, lab-measured current profiles for both UDDS and US06 were deliberately degraded with a constant +30 mA bias and additive Gaussian noise ($\sigma = 20$ mA) — representative of automotive-grade current sensor error. This is precisely the mechanism that causes an open-loop Coulomb Counting estimator to drift: sensor errors integrate over time instead of averaging out.

4.2 Coulomb Counting Baseline

The baseline estimator integrates measured current directly, with no correction mechanism:

$$SOC_{k+1} = SOC_k - \frac{T_s}{Q_{rated}} I_k^{meas}$$

Open-loop; error accumulates monotonically with sensor bias.

4.3 Extended Kalman Filter Formulation

The EKF treats SOC and the RC-branch voltage V1 as a two-state system, propagated forward using the current measurement, then corrected using the terminal voltage residual. Each cycle of the filter has four conceptual stages:

- **Prediction** — project SOC and V1 forward one timestep using the current measurement and the battery model, before any voltage correction is applied.
- **Innovation** — compute the residual between the terminal voltage the model predicted and the terminal voltage actually measured; this residual is the only new information entering the filter each step.
- **Kalman gain** — decide how much to trust that residual versus the model's own prediction, based on the current uncertainty (covariance) of each.
- **Correction** — apply the weighted residual to update the SOC and V1 estimates, and shrink the uncertainty accordingly before the next prediction.

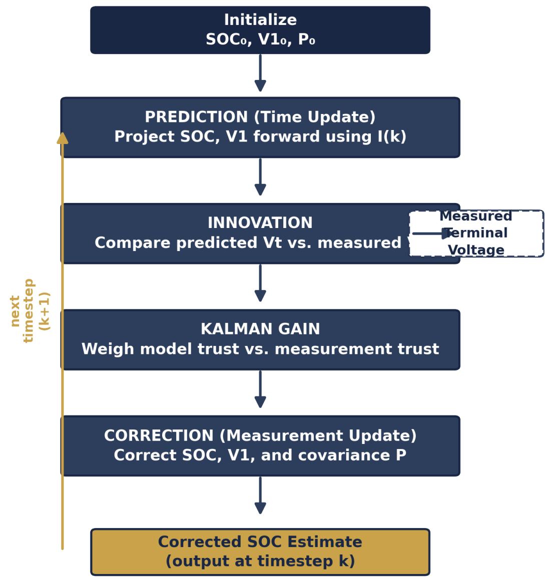

Figure 5. EKF cycle: prediction, innovation, gain computation, and correction, repeated every timestep.

State propagation (prediction step):

$$SOC_{k+1} = SOC_k - \frac{\eta T_s}{Q_{rated}} I_k$$

SOC evolves via current integration (η = coulombic efficiency).

$$V_{1,k+1} = e^{-T_s/(R_1 C_1)} V_{1,k} + R_1 \left(1 - e^{-T_s/(R_1 C_1)}\right) I_k$$

RC-branch voltage relaxes with time constant τ = R1·C1.

Measurement model:

$$V_{t,k} = OCV(SOC_k) - V_{1,k} - I_k R_0$$

Predicted terminal voltage from the 1RC model.

Linearization (Jacobian of OCV about the current SOC estimate):

$$H_k = \left[\frac{\partial OCV}{\partial SOC}(SOC_k) \quad -1 \right]$$

Since OCV(SOC) is nonlinear, EKF linearizes it at each step.

Correction step:

$$K_k = P_k^- H_k^T \left(H_k P_k^- H_k^T + R \right)^{-1}$$

Kalman gain balances model trust vs. measurement trust.

$$\hat{x}_k = \hat{x}_k^- + K_k \left(V_{t,k}^{meas} - V_{t,k}^{pred} \right)$$

State correction using the voltage innovation (residual).

$$P_k = (I - K_k H_k)\, P_k^-$$

Error covariance is updated after each correction.

This voltage-feedback loop is what Coulomb Counting lacks: every timestep, the EKF checks its own prediction against a real, independent measurement (terminal voltage) and corrects itself accordingly — which is why it does not drift indefinitely the way open-loop integration does.

05 Simulink Implementation

The estimation methodology above is implemented as a single Simulink model that runs Coulomb Counting and the EKF side by side against the same noisy current/voltage inputs, so both estimators see identical data each timestep.

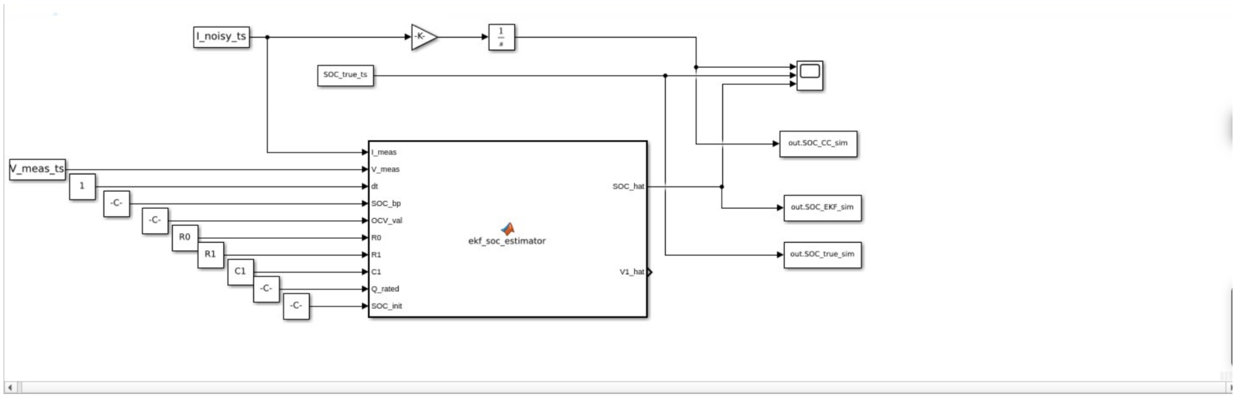

Figure 6. Simulink model architecture. Noisy current (I_noisy_ts) is integrated for the Coulomb Counting path (gain + discrete integrator, top) and fed alongside measured voltage (V_meas_ts) into the ekf_soc_estimator MATLAB Function block, which also receives the fixed battery parameters (R0, R1, C1, Q_rated) and the OCV breakpoint/value tables (SOC_bp, OCV_val). The block outputs SOC_hat and V1_hat; SOC_true_ts, the Coulomb Counting result, and the EKF result are logged together to a Scope and to workspace variables (out.SOC_CC_sim, out.SOC_EKF_sim, out.SOC_true_sim) for RMSE post-processing.

Running both estimators inside one model — rather than separate scripts — removes any risk of the two seeing slightly different current profiles, which keeps the RMSE comparison in Section 7 a clean test of the estimation algorithm alone.

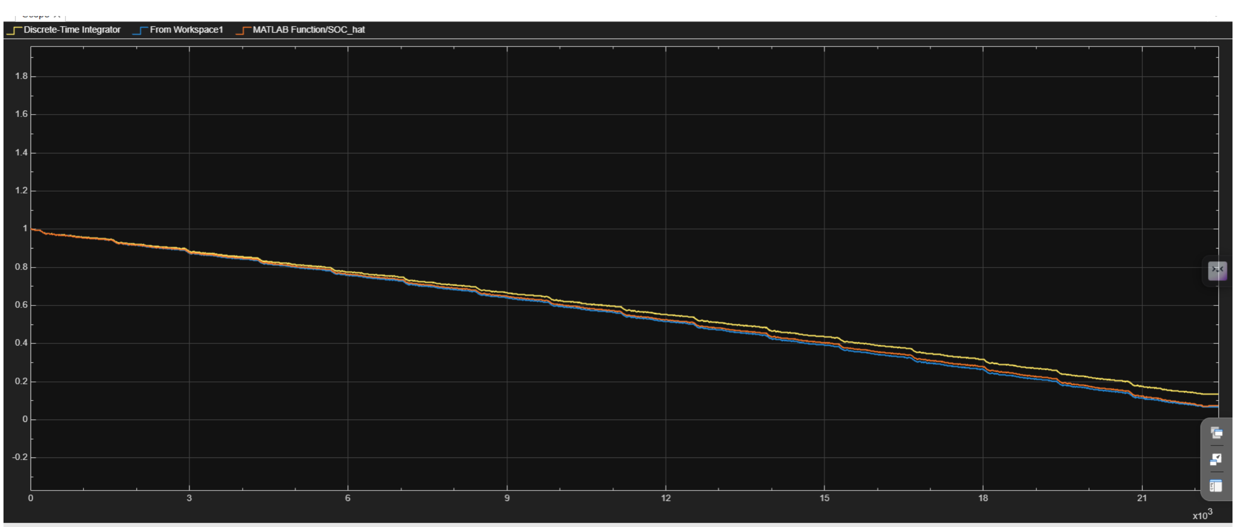

Figure 7. Live Scope output during simulation: Coulomb Counting (Discrete-Time Integrator, yellow), the logged true SOC reference (From Workspace1, blue), and the EKF estimate (MATLAB Function/SOC_hat, orange) tracked together over the full run, all three converging from SOC = 1.0 toward full discharge.

06 Filter Tuning and Robustness Validation

Tuning the process noise covariance (Q) and measurement noise covariance (R) required care, because the two drive cycles pull the filter in different directions: UDDS rewards trusting the voltage measurement heavily, while US06's fast transients punish that same trust with instability.

An initial grid search restricted to UDDS alone found parameters that achieved a highly accurate 0.51% RMSE — but the same tuning diverged to 4.44% RMSE when applied to US06. The filter had overfit to UDDS's gentle current profile and became unstable under US06's rapid transients.

To correct this, a **joint-optimization routine** was run to minimize the *worst-case* RMSE across both cycles simultaneously, rather than tuning against UDDS in isolation. This forced an explicit tradeoff: Q was tightened to diag([1e-11, 1e-6]) and R set to 10.0, shifting the filter to trust its internal model more and the noisy voltage measurement less. This sacrifices a small amount of best-case UDDS accuracy in exchange for eliminating the US06 instability — a deliberate robustness-over-peak-accuracy design choice appropriate for a production BMS that cannot know in advance what driving style it will see.

Tuning Strategy	UDDS RMSE	US06 RMSE	Outcome
UDDS-only grid search	0.51%	4.44%	Overfit — unstable on US06
Joint worst-case optimization	**0.93%**	**0.95%**	**Robust across both cycles**

Table 2. Effect of single-cycle vs. joint tuning on filter stability.

07 Results

Drive Cycle	Algorithm	RMSE	Δ vs. EKF
UDDS (Gentle)	Coulomb Counting	3.81%	—
UDDS (Gentle)	**Extended Kalman Filter**	**0.93%**	**4.1x lower**
US06 (Aggressive)	Coulomb Counting	0.79%	—
US06 (Aggressive)	**Extended Kalman Filter**	**0.95%**	**Consistent, bounded**

Table 3. RMSE comparison across drive cycles and algorithms.

Interpretation

On UDDS, the EKF cuts error roughly 4x relative to Coulomb Counting: the cycle is long and gentle enough that sensor drift has time to accumulate, and the EKF's voltage-based correction repeatedly pulls the estimate back toward truth. On US06, Coulomb Counting narrowly outperforms the EKF (0.79% vs. 0.95%) — and this is expected, not a filter shortcoming: US06 is short, and its current profile oscillates with near-zero net bias over the test window, so drift has not yet had time to surface. Coulomb Counting looks artificially strong here precisely because the test window is too short to expose its structural weakness.

The more diagnostic comparison is **consistency**. Coulomb Counting's error swings from 0.79% to 3.81% — a 4.8x variation — purely as a function of which cycle it happens to see, while the EKF stays tightly bounded between 0.93% and 0.95% regardless of driving style. A real vehicle does not get to choose its drive cycle in advance; a BMS needs SOC accuracy that holds up across unknown, sustained, real-world driving — exactly where Coulomb Counting's drift would eventually surface even on US06-like driving if extended from minutes to hours.

08 Computational Environment

Reproducibility details for the MATLAB/Simulink implementation:

Item	Detail
MATLAB / Simulink version	R2026a
Solver configuration	Mixed discrete–continuous (discrete state estimator, continuous battery plant)
Sample time (Ts)	≈ 1 s (matched to UDDS/US06 logged data interval)
Simulation duration	Full length of each drive cycle (UDDS and US06, run independently)

Table 4. Simulation environment and configuration. Confirm exact solver name and Ts against your model's Configuration Parameters before final submission.

09 Limitations and Future Work

- Results are specific to the Panasonic 18650PF cell chemistry and form factor; generalizing to other cells would require re-running HPPC extraction for R0, R1, C1, and the OCV–SOC table.
- The US06 test window is short relative to typical drift timescales; a longer or repeated US06 sequence would test whether Coulomb Counting's apparent edge holds up under sustained aggressive driving.
- Temperature effects on internal resistance and OCV were not modeled — a standard next step for production-grade BMS development.
- Planned extensions: adaptive (fading-memory) Q/R tuning, validation against additional real-world cycles such as WLTP and HWFET, and pack-level SOC estimation accounting for cell-to-cell variation.

10 Conclusion

The jointly-tuned EKF achieves what Coulomb Counting structurally cannot: stable, bounded SOC error across drive cycles of very different character, by continuously correcting for sensor drift using independent voltage feedback rather than relying purely on open-loop current integration. This robustness — not peak accuracy on any single, favorable cycle — is the property that matters for a BMS deployed in unpredictable, real-world conditions. The same modeling and joint-tuning approach developed here extends naturally to pack-level estimation, adaptive noise-covariance tuning, and additional validation cycles as next steps.

Full MATLAB/Simulink implementation, tuning scripts, and simulation data available at github.com/vidhanshu-tenwalia